

First name:

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SOLUTIONS

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Problem 1. Use the appropriate Lamé coefficients to compute the Laplacian of the scalar function $f(x, y, z) = x^2 + y^2 + z^2$ in both Cylindrical and Spherical coordinates. You must compute all the required partial derivatives.

For generic coordinates (q_1, q_2, q_3) :

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right].$$

For cylindrical coordinates:

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

For spherical coordinates:

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

[5 + 5 = 10 marks]

Solution.

In cylindrical coordinates: $r^2 = x^2 + y^2$

Thus $f = x^2 + y^2 + z^2 = r^2 + z^2$, hence $\frac{\partial f}{\partial \theta} = 0$

Therefore

$$\begin{aligned} \nabla^2 f &= \frac{1}{(1)(r)(1)} \left[\frac{\partial}{\partial r} \left(\frac{(r)(1)}{(1)} \frac{\partial (r^2 + z^2)}{\partial r} \right) + 0 + \frac{\partial}{\partial z} \left(\frac{(1)(r)}{(1)} \frac{\partial (r^2 + z^2)}{\partial z} \right) \right] \\ &= \frac{1}{r} \left[\frac{\partial}{\partial r} (r(2r)) + \frac{\partial}{\partial z} (r(2z)) \right] \\ &= \frac{1}{r} \left[\frac{\partial}{\partial r} (2r^2) + \frac{\partial}{\partial z} (2rz) \right] = \frac{1}{r} [4r + 2r] = \frac{1}{r} (6r) \\ &\Rightarrow \nabla^2 f = 6 \end{aligned}$$

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Extra Page: Must submit even if unused.

In Spherical Coordinates:

$$\rho^2 = x^2 + y^2 + z^2$$

Thus $f = x^2 + y^2 + z^2 = \rho^2$

So $\frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial \phi} = 0$

Therefore:

$$\begin{aligned} \nabla^2 f &= \frac{1}{(1)(\rho \sin \phi)(\rho)} \left[\frac{\partial}{\partial \rho} \left(\frac{\rho \sin \phi}{(1)} \frac{\partial}{\partial \rho} \rho^2 \right) + 0 + 0 \right] \\ &= \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} \left(\rho^2 \sin \phi \cdot (2\rho) \right) \right] \\ &= \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} \left(2\rho^3 \sin \phi \right) \right] = \frac{1}{\rho^2 \sin \phi} \left(6\rho^2 \sin \phi \right) \\ \Rightarrow \nabla^2 f &= 6 \end{aligned}$$

□